



DEPARTMENT OF POLITICAL SCIENCE

GOVERNANCE CAPACITY AND THE CLIMATE ADAPTATION IMPLEMENTATION GAP IN THE EUROPEAN UNION

Explaining Between- and Within-Country Variation in Climate Adaptation Performance Through Panel Data Analysis

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Abstract

The climate adaptation implementation gap has become an increasingly important concern in climate governance research, particularly within the European Union (EU), where countries face similar policy commitments but display substantial variation in adaptation performance. While existing literature often links stronger governance capacity directly to improved climate adaptation outcomes, these relationships are not always examined in an integrated framework that jointly accounts for structural and developmental conditions alongside governance effects to reduce the adaptation implementation gap. This thesis addresses this gap by examining how governance capacity shapes climate adaptation performance across and within 27 EU member states over the period 1995-2023. Using panel data analysis, the study employs fixed-effects (FE) and random-effects (RE) models to distinguish between within-country changes over time and between-country differences. Climate adaptation performance is measured using the inverse of the ND-GAIN vulnerability index, while governance capacity is operationalized using the World Bank Government Effectiveness indicator from the Quality of Government (QoG) dataset. The findings show that governance capacity is positively associated with climate adaptation performance in baseline models. However, this relationship weakens and becomes statistically insignificant once structural socioeconomic factors are included. The results further indicate that most variation in adaptation performance is driven by persistent differences between countries rather than changes within countries over time, with fixed-effects estimates revealing limited temporal variation. Importantly, the study finds that governance capacity influences climate adaptation primarily as part of broader structural and developmental conditions rather than as an independent short-term driver of adaptation performance.

Key Words: *Climate adaptation; Governance capacity; European Union; Adaptation implementation gap*

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1.Introduction:

Climate change is one of the most pressing global challenges, affecting societies, economies, and ecosystems. In recent years, climate policy has increasingly emphasized not only mitigation but also adaptation, which refers to the process through which countries adjust their systems to reduce vulnerability to climate-related risks such as floods, heatwaves, and extreme weather events (IPCC, 2014, p. 5). Within the European Union (EU), adaptation has become a central component of climate governance. Most EU countries have adopted national adaptation strategies and action plans. However, despite these strong policy commitments, there is a persistent gap between planned adaptation measures and actual outcomes. This is often referred to as the adaptation implementation gap (Intergovernmental Panel on Climate Change, 2014; United Nations Environment Programme, 2021). The adaptation implementation gap can be understood as the difference between adaptation needs and actual adaptation outcomes or capacity (Biesbroek et al., 2015). In other words, countries may adopt policies but fail to implement them effectively. This gap is clearly visible across EU countries. For example, Countries such as Sweden, Denmark, and the Netherlands consistently score high on climate adaptation readiness, while countries such as Bulgaria, Romania, and Croatia tend to show lower adaptation performance, reflecting differences in institutional and governance capacity (Notre Dame Global Adaptation Initiative [ND-GAIN], 2024; Moser & Ekstrom, 2010). This suggests that formal policy commitment alone is not sufficient to ensure successful adaptation outcomes. While existing research has examined the role of economic resources, vulnerability, technology, and institutional factors in shaping climate adaptation outcomes, less attention has been given to how governance capacity influences actual adaptation performance within a systematic cross-national framework (Dupuis & Biesbroek, 2013). In particular, it remains unclear whether the relationship between governance capacity and adaptation performance reflects an independent governance effect or broader structural differences across countries and over time. Building on this background, the central research question guiding this thesis is: *How and to what degree does governance capacity shape variation in climate adaptation performance across EU countries, and is this relationship consistent both across countries and within them over time?*

2. Literature Review: Defining key concepts

This section reviews existing literature on the role of governance capacity in shaping climate policy adaptation and implementation. It also clarifies key concepts related to climate policy that are relevant to this study. Building on this, the section identifies important gaps in the literature, particularly the persistent variances between strong policy commitments and actual adaptation outcomes. Despite the presence of well-developed climate strategies in many countries, implementation often falls short in practice. Understanding this Adaptation-implementation gap provides the central motivation for this thesis.

2.1 Defining Climate Adaptation

Climate adaptation refers to the process through which ecological, social, and economic systems adjust to actual or expected climate change in order to reduce harm or take advantage of potential benefits (Adger et al., 2005; IPCC, 2014). Adaptation includes both planned government policies and broader societal responses. However, in this study, the focus is specifically on the implementation¹ of planned adaptation measures by governments, refers to the process through which formally adopted climate adaptation policies are translated into concrete actions, programs, and outcomes in practice (Intergovernmental Panel on Climate Change, 2014, p. 5; Biesbroek, Robbert et al., 2015). Yet, an important distinction must be made between policy adoption and policy implementation. While many countries formally adopt climate adaptation strategies, the effectiveness of these policies depends on how well they are implemented in practice. This creates the need to examine the adaptation implementation gap, which refers to the difference between policy intentions and actual outcomes (Pressman & Wildavsky; Biesbroek et al., 2015).

2.2 What Adaptation Implementation Gap is:

The adaptation implementation gap refers to the difference between what countries aim to achieve and what they actually achieve in terms of adaptation (Biesbroek et al., 2015). This gap can emerge due to financial constraints, lack of coordination, weak institutions, or political barriers. Dupuis and Biesbroek (2013) also highlight the “dependent variable problem” in adaptation research, arguing that adaptation implementation and outcomes are difficult to measure consistently across countries

¹In the context of climate change policy, implementation refers to the process of translating formally adopted adaptation strategies into concrete actions and measurable outcomes (Intergovernmental Panel on Climate Change, 2014, p. 5; Biesbroek, Robbert et al., 2015).

due to differences in indicators, policy contexts, and evaluation methods. This makes it even more important to carefully operationalize adaptation in empirical studies. This study addresses this by using the ND-GAIN Adaptation Index as a comparative measure of adaptation performance, while clearly distinguishing it from governance capacity and critically discussing its limitations. For example- The 2003 and 2022 European heatwaves demonstrate significant variation in adaptation implementation across EU states. France has developed extensive heat-health action plans after the 2003 crisis, yet studies show persistent implementation gaps at the municipal level, particularly in coordination between health services and local authorities (Dupuis & Biesbroek, 2013). Similarly, in Southern European countries such as Spain and Italy, urban heat adaptation strategies exist in national planning documents, but their implementation is uneven due to financial constraints and fragmented governance structures, especially in smaller municipalities (EEA, 2021). Another example like a broader structural pattern in the EU shows an adaptation implementation gap between Western and Eastern member states. Western European countries such as Sweden, the Netherlands, and Germany generally exhibit more advanced adaptation planning and monitoring systems, yet still face implementation challenges at the local level due to coordination problems (Biesbroek et al., 2015). In contrast, several Eastern European countries have developed national adaptation strategies only recently, and their implementation is constrained by limited financial resources, weaker institutions, and lower administrative capacity (EEA, 2022). This illustrates the “dependent variable problem” in adaptation research, where difficulties in measuring adaptation lead to significant variation in reported adaptation outcomes and implementation processes, even under similar EU policy frameworks (Dupuis & Biesbroek, 2013). It also highlights the challenges of multi-level governance, as nationally adopted adaptation strategies are often implemented unevenly across regional and local levels.

2.3 Governance and Administrative Capacity in Policy Implementation

While climate adaptation research increasingly recognizes the importance of institutions, it has not fully integrated insights from public administration and governance research. In particular, the concept of administrative capacity, widely regarded as essential for effective policy implementation in other policy areas, remains underexplored in climate adaptation studies. At the same time, governance and state capacity are contested and multidimensional concepts, as scholars define them differently depending on institutional and policy contexts (Fukuyama, 2013, 2016). As Fukuyama (2013, p. 3) argues, “there is very little agreement on what constitutes high-quality government.”

Broadly, administrative capacity refers to the ability of public institutions to design, coordinate, and effectively implement policies. It includes bureaucratic efficiency, resource availability, policy expertise, coordination mechanisms, and the ability to enforce rules and deliver public services (Bachtler et al., 2014; Mendez & Bachtler, 2022). However, governance capacity extends beyond technical efficiency alone. Research on quality of government emphasizes the importance of impartiality, accountability, and low levels of corruption in public institutions (Rothstein & Teorell, 2008). High-quality government institutions are characterized by the fair and impartial implementation of public policies rather than favoritism, corruption, or political clientelism (Rothstein & Teorell, 2008). Similarly, Fukuyama (2013, 2016) argues that governance capacity should be understood not only in terms of state effectiveness but also through the quality and integrity of institutions. These distinctions are particularly important in the context of climate adaptation because adaptation outcomes depend not only on whether governments adopt policies, but also on whether institutions possess the capacity to implement them effectively. Research on EU cohesion policy demonstrates that differences in administrative capacity strongly influence policy outcomes across member states. Bachtler, Mendez, and Oraže (2014) show that weaker institutional structures in Central and Eastern Europe often reduce the effectiveness of cohesion policy implementation. Likewise, Mendez and Bachtler (2022) find that higher quality of government is associated with stronger policy compliance and better implementation outcomes across EU regions. Quantitative studies further support this relationship. Incaltarau, Pascariu, and Surubaru (2020) demonstrate that administrative capacity and political governance are key factors explaining differences in EU fund absorption among member states. Tosun (2014) similarly finds that bureaucratic efficiency affects regional policy implementation, while Terracciano and Graziano (2016) highlight the importance of subnational administrative structures in decentralized governance systems such as Italy. More recently, Bachtrögl (2024) argues that administrative capacity is essential not only for absorbing resources but also for achieving intended policy goals.

In addition to governance and administrative capacity, previous research suggests that climate adaptation performance is also shaped by broader economic, political, and demographic conditions. Economic development is frequently associated with stronger fiscal capacity, technological resources, and institutional infrastructure for climate adaptation (IPCC, 2014; UNEP, 2021). Political context may also influence policy effectiveness, as democratic institutions can strengthen accountability, transparency, and public participation in environmental governance (Povitkina, 2018),

while corruption and weak institutional integrity may reduce implementation effectiveness and public trust (Rothstein & Teorell, 2008; Sundström et al., 2026). Furthermore, demographic conditions such as population size and urbanization may increase governance complexity and infrastructure pressure, thereby affecting adaptation capacity and implementation processes (Eakin et al., 2018; Seto et al., 2014). These factors are therefore important to consider alongside governance capacity when analyzing variation in climate adaptation performance across EU countries.

3. Previous Research Gaps and Research Questions:

A critical review of the existing literature reveals several important gaps in climate adaptation research. Much of the literature remains dominated by qualitative and case-based approaches, limiting the ability to generalize findings across countries and conduct systematic comparative analysis of adaptation performance (Dupuis & Biesbroek, 2013; Moyo et al., 2026). Dupuis and Biesbroek (2013) argue that adaptation research faces a “dependent variable problem” because many studies rely on context-specific cases that are difficult to compare systematically across countries. Similarly, Görbig et al. (2026) examine the adaptation planning-implementation gap through evidence from European cities and highlight the fragmented nature of adaptation research across local contexts. Moyo et al. (2026) further show that research on multi-level governance and adaptation continues to rely heavily on qualitative governance analysis rather than cross-national quantitative approaches.

Recent research highlights a persistent gap between climate adaptation planning and implementation. Although many governments formally adopt adaptation strategies, they often struggle to translate these commitments into effective action due to governance, coordination, and resource-related barriers (Görbig et al., 2026; Roest et al., 2025). Studies on multi-level governance show that weak coordination between national, regional, and local institutions frequently constrains implementation processes (Moyo et al., 2026). Similarly, limited financial and administrative resources reduce the adaptive capacity of local governments, particularly in smaller municipalities (Fila et al., 2024). The European Environment Agency (2025) likewise reports that, despite progress in adaptation planning across Europe, major implementation challenges remain, especially at the local level.

At the same time, much of the adaptation literature continues to focus on policy adoption rather than actual implementation outcomes, often evaluating whether adaptation strategies exist without examining whether they produce measurable improvements in resilience (Biesbroek et al., 2015).

Although institutional factors are increasingly acknowledged, governance and administrative capacity have not been systematically analyzed as central determinants of adaptation performance. This is a significant limitation because extensive EU cohesion policy research demonstrates that administrative capacity strongly shapes policy effectiveness and implementation outcomes across member states (Bachtler et al., 2014; Incaltarau et al., 2020; Mendez & Bachtler, 2022; Bachtrögler, 2024).

In summary, these limitations point to a broader gap in the literature such as- the lack of systematic quantitative research examining how governance and administrative capacity influence climate adaptation implementation outcomes across and within EU countries. Existing adaptation research often treats the European Union as institutionally homogeneous, despite substantial variation in governance quality and implementation capacity among member states. As a result, adaptation studies may overlook an important explanatory factor behind differences in adaptation performance.

This study addresses this gap by conducting a comparative cross-national analysis of governance capacity and climate adaptation performance across EU member states. Unlike much of the existing literature, which focuses mainly on policy adoption and qualitative case studies, this study examines actual adaptation outcomes through a quantitative comparative approach. By operationalizing adaptation through the ND-GAIN Adaptation Index while critically acknowledging its limitations, the study contributes to a more empirically grounded understanding of why climate adaptation performance varies across EU countries and how governance capacity shapes climate policy effectiveness. Thus, these gaps in the literature motivate the central focus of this thesis. Existing research demonstrates significant variation in climate adaptation implementation across EU countries, yet the role of governance and administrative capacity in explaining this variation remains insufficiently explored. Therefore, the following research question and sub-questions are developed to examine this relationship through quantitative analysis and discussion of the empirical findings. The key research question of the study is *Does stronger governance capacity reduce the climate adaptation implementation gap within and across EU countries?* Under the key research question, we also have some sub-research questions:

- i) *Do countries with higher governance capacity exhibit better adaptation performance?* ii) *Do improvements in governance capacity within a country over time lead to measurable improvements in adaptation performance and reduce adaptation implementation gap?* iii) *To what extent is*

variation in adaptation performance explained by differences between countries versus changes within countries over time?

4. Theoretical Framework and Hypothesis

The theoretical foundation of this study is grounded in policy implementation theory, governance theory, and state capacity theory, all of which emphasize the importance of institutional capacity in translating policy goals into effective outcomes. Together, these perspectives help explain why some countries are more successful than others in implementing climate adaptation policies.

The starting point of this framework comes from policy implementation theory. Pressman and Wildavsky (1973) argue that policy implementation is not a simple or linear process but a complex chain of decisions involving multiple actors, institutions, and administrative stages where failure can occur at many points. According to their argument, successful policy outcomes depend not only on the formal adoption of policies but also on governments' ability to coordinate and implement them effectively in practice. This perspective is particularly relevant to climate adaptation because adaptation policies usually require long-term coordination across ministries, local governments, environmental agencies, and other public institutions. As a result, even when governments formally adopt adaptation strategies, implementation gaps may emerge if coordination mechanisms and administrative systems are weak (Pressman & Wildavsky, 1973).

Building on this perspective, governance theory emphasizes the role of effective institutions in policymaking and implementation. Governance scholars argue that effective governance refers to the ability of governments to design, coordinate, and enforce policies efficiently and transparently (Fukuyama, 2013). Governance capacity therefore includes the quality of public services, bureaucratic competence, accountability, and the ability of institutions to implement policies consistently. In this study, governance capacity is primarily captured through government effectiveness, which reflects administrative competence and policy implementation capacity. Research on quality of government further highlights that effective governance also depends on impartiality and low levels of corruption in public institutions (Rothstein & Teorell, 2008). Weak governance structures may reduce policy effectiveness by creating inefficiency, fragmented coordination, and limited institutional trust. In climate adaptation, these governance challenges are

particularly important because adaptation policies require sustained coordination, long-term planning, and continuous implementation across multiple sectors and levels of government.

State capacity theory further strengthens this argument by focusing on the ability of states to mobilize resources and carry out public policies effectively. Grindle (1996) argues that institutional capacity refers to the ability of public organizations to perform tasks effectively, efficiently, and sustainably. Similarly, Fukuyama (2013, p. 353) defines state capacity as “*the ability of states to plan and execute policies and to enforce laws cleanly and transparently.*” These insights are highly relevant for climate adaptation because adaptation implementation depends heavily on governments’ ability to allocate resources, coordinate institutions, enforce environmental regulations, and sustain policy commitments over time. Adaptation policies are also governance-intensive because they involve uncertainty, long-term investment, and multi-level implementation challenges (Adger et al., 2005). Where state capacity is weak, implementation gaps are more likely to emerge due to limited administrative capability and ineffective coordination (Klein et al., 2014; Eakin et al., 2018).

An additional dimension of governance capacity relates to administrative systems such as public procurement, which influence transparency, accountability, and efficiency in policy delivery. Weak procurement systems may contribute to inefficiency and misallocation of resources, thereby reducing implementation effectiveness (Thai, 2001; Gordon, 2009). Combinedly, these theoretical perspectives suggest that climate adaptation is a governance-intensive policy domain in which institutional quality and administrative capacity play a decisive role in shaping implementation outcomes.

Beside this, although governance capacity and adaptation performance are theoretically related, they remain conceptually distinct. Governance capacity refers to the institutional and administrative ability of the state to design and implement policies effectively, whereas adaptation performance reflects actual outcomes in reducing climate vulnerability (Savoia & Sen, 2015; Dupuis & Biesbroek, 2013). A country may possess relatively strong administrative capacity but still experience weaker adaptation outcomes due to geographic exposure, economic structure, political priorities, or environmental conditions. Conversely, some countries may achieve comparatively stronger adaptation outcomes despite more limited governance capacity, reflecting the multidimensional and context-dependent nature of climate adaptation (Adger et al., 2005; Biesbroek et al., 2015). Strong governance capacity therefore increases the likelihood that adaptation policies are implemented

effectively, which leads to the formulation of the first hypothesis of this thesis, tested through a cross-country analytical framework.

H1: Countries with higher governance capacity, measured through government effectiveness, are expected to exhibit higher levels of climate adaptation performance and lower climate adaptation implementation gaps across EU countries over time.

Cingolani (2013) defines state capacity as the ability of states to achieve policy objectives and implement decisions effectively. She argues that state capacity is multidimensional, including administrative capacity, bureaucratic quality, implementation ability, and institutional effectiveness. Importantly, Cingolani emphasizes that state capacity should not be understood merely as the existence of formal institutions, but as governments' practical ability to translate political goals into real-world outcomes. This distinction is highly relevant to climate adaptation because many EU countries have adopted similar adaptation strategies, yet their implementation outcomes vary significantly due to differences in governance and administrative effectiveness. Cingolani (2013) further identifies government effectiveness as a useful indicator of state capacity because it captures bureaucratic competence, policy implementation ability, and the quality of public services. This directly supports the operationalization used in this study, where government effectiveness is used as the main indicator of governance capacity.

The concept of governance quality is further developed by Rothstein and Teorell (2008), who define quality of government as the impartiality in the exercise of public authority. They argue that effective institutions are characterized by low corruption, impartiality, and consistent policy implementation, which strengthen public trust and policy credibility. This is particularly relevant for climate adaptation because adaptation policies require long-term coordination across multiple institutions and administrative levels.

The broader state capacity literature further supports this relationship. Savoia and Sen (2015) argue that state capacity is fundamental for effective governance and identify administrative, legal, fiscal, and infrastructural capacity as key dimensions. Similarly, Meckling and Nahm (2018) emphasize that climate policy success depends not only on political ambition but also on "process capacity," meaning the institutional ability to coordinate actors and sustain implementation over time. The climate adaptation literature also supports this argument. Adger et al. (2005), Klein et al. (2014), and Eakin

et al. (2018) show that adaptation outcomes are strongly shaped by governance systems, institutional coordination, and implementation capacity.

Empirical evidence from country cases further illustrates this relationship. The Netherlands demonstrates how long-term improvements in governance capacity can strengthen adaptation outcomes. Following severe flood risks, the Dutch government developed integrated water governance systems, strengthened coordination between national and regional authorities, and established long-term programs such as the Delta Programme. Similarly, Germany improved adaptation implementation over time by strengthening coordination across federal and regional levels through national adaptation strategies and monitoring systems. These cases support Meckling and Nahm's (2018) argument that sustained "process capacity" and institutional coordination are essential for effective climate policy implementation.

Therefore, the theoretical framework suggests that climate adaptation implementation is strongly dependent on governance capacity. Countries with stronger institutional quality and administrative effectiveness are expected to coordinate adaptation policies more successfully and achieve better implementation outcomes, while weaker governance structures are more likely to generate persistent implementation gaps.

In contrast, governments with weaker governance capacity are more likely to face implementation difficulties because of administrative inefficiency, fragmented institutions, weaker coordination mechanisms, and lower policy credibility. Based on this theoretical logic, the study develops the following second hypothesis which is tested using a fixed effects framework. The second hypothesis builds on the expectation that improvements in governance capacity within countries over time should contribute to stronger climate adaptation performance. The fixed-effects framework is particularly relevant because it focuses on within-country temporal changes rather than stable differences between countries. Theoretically, as governments strengthen administrative coordination, bureaucratic effectiveness, policy implementation capacity, and institutional quality over time, they should become more capable of translating adaptation strategies into practical outcomes. State capacity scholars argue that effective implementation depends on governments' ability to coordinate policies, manage resources, and sustain long-term institutional commitment (Cingolani, 2013; Savoia & Sen, 2015). Similarly, Meckling and Nahm (2018) emphasize that climate governance success depends on long-term "process capacity" and institutional coordination across sectors. Improvements in

governance systems over time may therefore strengthen adaptation planning, implementation, and monitoring mechanisms, contributing to better adaptation performance and reduced implementation gaps. Based on this theoretical logic, the study expects that increases in governance capacity within countries over time will be positively associated with improvements in climate adaptation performance and thereby establishes second hypothesis:

H2: Increases in government capacity within a country over time are expected to lead to improvements in climate adaptation performance and reduce adaptation implementation gap.

These hypotheses are important because they move beyond general discussions of climate policy ambition and focus specifically on adaptation-implementation outcomes and gaps. By examining both cross-country differences and within-country changes over time, the study aims to provide a clearer understanding of how governance capacity shapes the climate adaptation implementation gap within and across the EU context. The hypotheses also contribute to the broader state capacity and climate governance literature by empirically testing whether stronger and improving governance systems lead to more effective adaptation performance.

5. Methodological Approach and Design

5.1 Methodology:

This study uses a quantitative panel data design covering European Union (EU) countries between 1995 and 2023 in order to examine the relationship between governance capacity and climate adaptation performance over time. The period was selected primarily based on the availability of the ND-GAIN vulnerability dataset, which provides the main measure of climate adaptation performance across EU countries from 1995 onward. In addition, using a multi-decade panel allows the study to examine both cross-country differences and within-country changes over time, while maintaining consistency across the explanatory and control variables. A panel data approach is particularly suitable for this study because it allows the analysis to capture both differences between countries and changes within countries across time. This is important given that the research questions focus on both cross-national variation in adaptation performance and temporal changes in governance capacity within countries. The empirical analysis employs both Random Effects (RE) and Fixed

Effects (FE) regression models. The use of both models reflects the dual analytical focus of the study and allows different dimensions of variation to be examined systematically.

The Random Effects model² is primarily used to analyze cross-country variation in climate adaptation performance. This model estimates how structural differences in governance capacity across EU countries are associated with differences in adaptation outcomes over time. Because the RE model incorporates both between-country and within-country variation, it is particularly relevant for addressing the first research question concerning whether countries with higher governance capacity exhibit better adaptation performance. Accordingly, the first hypothesis is tested through the RE specification: *H1: Countries with higher governance capacity tend to exhibit higher levels of climate adaptation performance and lower climate adaptation implementation gaps across EU countries over time.* Within this framework, countries with consistently stronger institutional quality and government effectiveness are expected to demonstrate higher levels of adaptation performance across the observed period.

The Fixed Effects model³ is employed as a complementary analytical strategy to examine within-country variation over time. Unlike the RE model, FE controls for all time-invariant country-specific characteristics, such as geography, historical institutional structures, or long-term cultural factors, by focusing only on changes occurring within each country over time. This approach allows the study to assess whether improvements in governance capacity within a given country are associated with corresponding improvements in climate adaptation performance. Therefore, the FE model is directly linked to the second hypothesis: *H2: Increases in governance capacity within a country over time lead to improvements in climate adaptation performance and reduce adaptation implementation gap.*

By examining temporal changes within countries, the FE model strengthens the internal validity of the analysis and reduces the risk that estimated relationships are driven by unobserved country-specific characteristics. To further strengthen the analysis, year fixed effects are included in the regression models in order to account for common temporal shocks and broader EU-wide trends that

² Random-effects models incorporate both within-country and between-country variation under the assumption that unobserved country-specific effects are not systematically correlated with the explanatory variables (Baltagi, 2021).

³ Fixed-effects models estimate relationships using within-country variation over time by controlling for all time-invariant country-specific characteristics (Wooldridge, 2013).

may affect all countries simultaneously, such as economic crises, major climate agreements, or changes in EU climate policy frameworks. In addition, clustered standard errors ⁴at the country level are used to address potential heteroskedasticity and serial correlation within panels.

The study also includes several control variables commonly associated with climate adaptation performance, including economic development (GDP per capita), democracy level, total population, and urban population rate. Including these variables helps isolate the relationship between governance capacity and adaptation performance by accounting for alternative structural and socio-economic explanations. Finally, the study employs several robustness checks to enhance the credibility of the findings. These include testing alternative model specifications examining lagged governance effects, and conducting subsample analyses across different time periods and country groups. In addition, diagnostic tests such as the Hausman test are used to assess model suitability and evaluate key panel-data assumptions.

Despite these methodological strengths, several limitations should be acknowledged. Although panel-data methods improve the ability to account for unobserved heterogeneity and temporal dynamics, the study remains observational in nature and therefore cannot establish definitive causal relationships. In addition, governance indicators such as government effectiveness serve as proxy measures and may not fully capture all dimensions of institutional capacity and policy implementation. Nevertheless, the combined use of fixed-effects and random-effects models, clustered standard errors, diagnostic testing, and multiple robustness checks strengthens the reliability and internal validity of the analysis.

AI tools were used to assist with troubleshooting, Stata code generation, clarification of statistical procedures and regression interpretation, and improvement of language and grammar. All research design decisions, analysis, interpretations, and conclusions were developed and reviewed independently by the author.

⁴ Clustered standard errors are commonly used in panel-data analysis to account for heteroskedasticity and serial correlation within panels, thereby improving the reliability of statistical inference (Cameron & Miller, 2015).

5.2: Data and material:

The sample includes all 27 EU countries, which ensures sufficient variation between high-capacity and lower-capacity states. In this study, climate adaptation performance is measured using the vulnerability score from the (Notre Dame Global Adaptation Initiative [ND-GAIN], 2024). This study also uses panel data from the Quality of Government Institute (QoG) Standard Dataset, which compiles data from multiple international sources, including the World Bank Worldwide Governance Indicators (WGI) for governance capacity, the World Development Indicators (WDI) for socioeconomic variables, and the Varieties of Democracy (V-Dem) dataset for democracy measures.

5.3: Operationalization of the variable

5.3.1: Dependent variable: Climate adaptation performance

Climate adaptation performance is the dependent variable in this study. It is operationalized using the vulnerability component of the ND-GAIN index developed by the University of Notre Dame Global Adaptation Initiative. The ND-GAIN framework is a widely used cross-national measure that assesses countries' exposure, sensitivity, and adaptive challenges related to climate change across six key sectors: food, water, health, ecosystems, infrastructure, and human habitat (Chen et al., 2015; ND-GAIN, 2023). The vulnerability index represents a composite measure derived from these sectoral indicators and captures the overall level of climate risk faced by a country. It combines multiple dimensions of climate vulnerability, including both biophysical exposure and socio-economic sensitivity, recognizing that countries facing similar climate hazards may differ in vulnerability depending on the resilience of their systems. As such, the index is neither purely objective nor purely subjective; rather, it integrates observable indicators with broader structural characteristics (Chen et al., 2015; ND-GAIN, 2023). Importantly, this study uses only the vulnerability component of the ND-GAIN index and excludes the readiness dimension. This is done to ensure a clear distinction between inputs and outputs, as the readiness component includes governance-related indicators that could overlap with the independent variable, governance capacity. By focusing on vulnerability, the analysis reduces potential endogeneity and strengthens the internal validity of the model. The vulnerability index is measured on a scale from 0 to 1, where higher values indicate greater climate vulnerability. For ease of interpretation and consistency with the theoretical framework, the dependent variable, Climate Adaptation Performance (*adapt_perf*), is constructed as the inverse of the vulnerability score using the transformation $\text{Climate Adaptation Performance} = 1 - \text{vulnerability}$. This transformation ensures that higher values indicate stronger adaptation

performance (and a lower adaptation implementation gap), making the results more intuitive to interpret in terms of improved climate adaptation outcomes. This transformation ensures that higher values of the variable correspond to better adaptation performance, which improves interpretability and consistency with the theoretical concept. In the original vulnerability measure, higher values indicate greater exposure and sensitivity to climate risks (i.e., worse outcomes). By reversing the scale, the new variable reflects the extent to which countries are able to reduce vulnerability and effectively adapt to climate change. For example, if a country has a vulnerability score of 0.70, its adaptation performance score becomes 0.30 ($1 - 0.70$), indicating relatively low adaptation capacity. Conversely, a country with a vulnerability score of 0.30 would have an adaptation performance score of 0.70, indicating stronger adaptation performance. This is done so that higher values consistently represent better adaptation performance, making the results easier to interpret.

While the ND-GAIN vulnerability index offers several analytical advantages for comparative research, it is also important to acknowledge alternative measures and the limitations associated with the selected operationalization. Alternative measures of climate adaptation and climate risk exist, including disaster-related indicators and the full ND-GAIN index combining vulnerability and readiness dimensions. However, disaster-based measures often capture short-term climate impacts rather than broader adaptation capacity, while the readiness component includes governance-related indicators that may overlap with the independent variable. In addition, vulnerability-based measures may partly favor countries located in less climate-exposed geographic regions, making direct cross-country comparisons more difficult because differences in environmental exposure are not determined solely by governance capacity or adaptation policy (Magnan et al., 2016). Focusing only on vulnerability therefore helps reduce potential endogeneity and maintain a clearer distinction between governance capacity and adaptation outcomes.

Despite these advantages, some limitations should be acknowledged. The ND-GAIN vulnerability index is a composite measure that simplifies a complex and multidimensional phenomenon into a single score. As a result, it may not fully capture differences in local adaptation processes or policy implementation quality. Despite these limitations, the ND-GAIN vulnerability index remains one of the most widely used and internationally comparable measures of climate vulnerability and adaptation-related performance, making it well suited for cross-national panel analysis (Chen et al., 2015; ND-GAIN, 2023).

5.3.2: Independent Variable: Government Effectiveness (governance capacity)

The main independent variable in this study is Government Effectiveness, which is used as a proxy for governance capacity. The variable is obtained from the World Bank Worldwide Governance Indicators (WGI) dataset and accessed through the Quality of Government Institute (QoG) Standard Dataset under the code *wbgi_gee*. The indicator measures perceptions of the quality of public services, the quality and professionalism of the bureaucracy, the competence and independence of civil servants, and the credibility of government commitment to policy implementation. The measure therefore focuses primarily on the institutional and administrative “inputs” required for governments to formulate and implement effective public policies (Codebook, The QOG Standard Dataset, 2026).⁵

Government Effectiveness is selected because it aligns closely with the theoretical argument of this thesis that stronger governance capacity increases the possibility of successful climate adaptation outcomes. Climate adaptation policies often require long-term planning, administrative coordination, resource allocation, regulatory enforcement, and implementation across multiple sectors and governance levels. As a result, countries with stronger state and bureaucratic capacity are generally expected to possess greater ability to formulate and implement effective adaptation measures (IPCC, 2014; Fukuyama, 2013; Meckling & Nahm, 2018). This relationship forms the basis of the study’s central hypothesis that higher governance capacity is associated with stronger adaptation performance. The use of Government Effectiveness is also supported by extensive comparative governance literature. Previous studies identify state capacity and bureaucratic quality as central determinants of policy implementation and institutional performance (Savoia & Sen, 2015; Cingolani, 2013). Similarly, climate governance research frequently emphasizes that institutional quality and administrative effectiveness shape states’ ability to respond to environmental challenges and implement adaptation strategies. By focusing on implementation-oriented institutional capacity, the indicator fits particularly well with this study’s emphasis on adaptation outcomes rather than merely policy commitments.

A major advantage of the Government Effectiveness indicator is its strong cross-national comparability and longitudinal coverage. The WGI dataset provides standardized measurements

⁵ Teorell, J., Sundström, A., Holmberg, S., Rothstein, B., Alvarado Pachon, N., Phiri, V. S., Chen, C., & Liu, Z. (2026). *The Quality of Government Standard Dataset* (Version Jan26) [Data set]. The Quality of Government Institute, University of Gothenburg. <https://doi.org/10.18157/qogstdjan26>

across countries and years, making it particularly suitable for panel analysis of EU member states over time. In addition, the indicator is widely used in comparative political science and public administration research, which increases the reliability and academic legitimacy of the operationalization.

Despite these strengths, few limitations should be acknowledged. First, Government Effectiveness remains a proxy measure and cannot fully capture all dimensions of governance capacity. Informal institutions, political leadership quality, coordination mechanisms, and issue-specific administrative capabilities may not be entirely reflected in the indicator. In addition, the measure is partly perception-based and relies substantially on expert assessments rather than purely objective administrative indicators, which may introduce subjective evaluation bias (Kaufmann et al., 2010). Second, EU countries generally exhibit relatively high governance standards, meaning that variation across cases may be more limited than in global comparisons.

To minimize these concerns, this study focuses specifically on the implementation-related dimension of governance capacity, which Government Effectiveness captures more directly than broader democracy or governance indicators. Furthermore, the use of panel data helps identify both cross-country differences and within-country variation over time, strengthening the analytical leverage of the study. The variable is also theoretically appropriate because it emphasizes the administrative and institutional capabilities necessary for translating climate adaptation policies into practical implementation outcomes.

5.3.3: Control variables

To account for alternative explanations and isolate the effect of governance capacity on climate adaptation performance, this study includes several control variables commonly used in comparative climate governance and public policy research. The inclusion of these controls is essential because climate adaptation outcomes are influenced not only by governance quality, but also by broader economic, political, and demographic conditions. Importantly, these variables may affect both governance capacity (independent variable) and adaptation performance (dependent variable), thereby creating potential confounding effects. Controlling for them improves internal validity by reducing the risk of omitted variable bias and spurious relationships.

Economic development. Economic development is controlled for using GDP per capita (current US dollars), obtained from the World Bank World Development Indicators through the Quality of

Government Institute (code: *wdi_gdpcapcur*). GDP per capita captures the level of economic resources available within a country and is widely used as an indicator of national wealth and development capacity (Teorell et al., 2026). Theoretically, GDP per capita is relevant for both governance capacity and climate adaptation performance. Wealthier countries tend to have greater fiscal space to invest in public administration, bureaucratic professionalism, and institutional development, which can strengthen governance capacity. On the other hand, higher income levels also provide governments with more financial and technological resources to invest in climate adaptation infrastructure, disaster preparedness, and resilience-building measures. Economic resources therefore shape both the quality of governance institutions and the ability to implement adaptation policies effectively (IPCC, 2014).

The use of GDP per capita also has important methodological strengths, including strong cross-national comparability and long time-series availability, making it suitable for panel analysis. However, GDP per capita does not capture differences in income distribution or how effectively economic resources are used for climate adaptation. High income does not automatically ensure strong adaptation performance, as institutional and governance factors like institutional quality and governance effectiveness also play a crucial role (Adger et al., 2005; IPCC, 2014). Despite these limitations, GDP per capita remains one of the most widely used and theoretically relevant indicators of development capacity in comparative political and environmental research.

Democracy level. Democracy is measured using the Electoral Democracy Index (code: *vdem_polyarchy*), developed by the Varieties of Democracy Project and accessed through the Quality of Government dataset (Teorell et al., 2026). The index captures core dimensions of electoral democracy, including free and fair elections, suffrage, freedom of expression, freedom of association, and access to alternative sources of information (Teorell et al., 2026).

This variable is included because democracy may influence both governance capacity and climate adaptation performance. Democratic systems can strengthen governance capacity by promoting accountability, reducing corruption, and improving responsiveness in public administration, also previous research suggests that democratic systems may create stronger incentives for governments to address environmental risks and respond to societal demands for climate protection. At the same time, democratic institutions may enhance adaptation outcomes by increasing policy responsiveness

to public concerns, strengthening environmental regulation, and improving transparency in decision-making processes (Povitkina, 2018).

A major strength of the Electoral Democracy Index is that it provides a multidimensional and highly detailed measure of democratic governance across countries and over time. However, it also has limitations. Some components rely on expert assessments and survey-based inputs, which may introduce measurement uncertainty and subjective interpretation (Varieties of Democracy Project, 2026)⁶. Furthermore, democracy and governance capacity are conceptually related but analytically distinct. Nevertheless, though democracy and governance quality are related, they measure different institutional dimensions. Government Effectiveness primarily measures bureaucratic and administrative implementation capacity, whereas the Electoral Democracy Index captures political and electoral institutions, reflecting the broader distinction between state capacity and democracy in governance research and highlighting the distinction between governance quality and regime type (Fukuyama, 2013; Rothstein & Teorell, 2008)

Population size. Total population is included as a control for country size and demographic scale. Population data are obtained from the World Bank World Development Indicators through the QoG dataset (code: *wdi_pop*) and measured as total midyear population estimates (Teorell et al., 2026). Population size is theoretically relevant because it may influence both governance capacity and climate adaptation performance. Population size is theoretically relevant because larger populations may place greater pressure on public infrastructure, natural resources, and administrative systems, potentially increasing exposure and vulnerability to climate-related risks. At the same time, larger countries may also have greater economic resources and more extensive institutional structures that can support adaptation investment and policy implementation (IPCC, 2014). Thus, population size may simultaneously shape the capacity of governance systems and the scale of adaptation challenges, making it a relevant confounder. The inclusion of population size helps account for structural differences between EU member states that could otherwise influence adaptation performance independently of governance quality. However, total population alone cannot capture population density, spatial distribution, or regional exposure to climate risks. Despite these limitations,

⁶ The Electoral Democracy Index combines expert-coded and survey-based indicators across dimensions such as electoral competition, freedom of expression, suffrage, and associational autonomy. As the index relies partly on expert judgments and aggregation methods, some measurement uncertainty may remain (Varieties of Democracy Project, 2026; QoG Codebook, *vdem_polyarchy*).

population size remains an important demographic control in cross-national comparative research because it reflects broader differences in size and governance demands across countries.

Urban population. Urban population is measured as the percentage of the population living in urban areas (code: *wdi_popurb*), based on data from the United Nations Population Division and accessed through the Quality of Government dataset (Teorell et al., 2026). This variable is used as a proxy for urbanization and reflects the degree of urban concentration within a country.

Urbanization is relevant for both governance capacity and climate adaptation performance. Higher levels of urban concentration may increase administrative demands and require stronger institutional coordination, potentially influencing governance capacity. At the same time, urban areas tend to have both advantages and risks for adaptation. They often benefit from better infrastructure, stronger service delivery, and greater access to technology, which can enhance adaptive capacity. However, urban regions are also more exposed to climate risks such as flooding, heat stress, environmental degradation, and infrastructure pressure (IPCC, 2014).

Including urban population therefore helps capture structural differences in settlement patterns that may jointly influence governance demands and adaptation outcomes. Urban population is measured as the percentage of the population living in urban areas and is used as a broad indicator of urbanization. A key advantage of this measure is its high cross-national comparability and long-term availability, which makes it suitable for panel analysis. However, urban population percentages cannot fully capture differences in urban planning quality, infrastructure resilience, or inequality within cities. To minimize this limitation, the variable is used primarily as a broad structural and demographic control rather than as a direct measure of urban adaptive capacity or urban governance quality. This allows the study to account for general differences in urban concentration across countries while avoiding overly specific assumptions about local resilience systems (IPCC, 2014).

Overall, these control variables improve the robustness of the empirical analysis by accounting for key structural, economic, political, and demographic factors that may influence both governance capacity and climate adaptation performance. Their inclusion reduces the risk of omitted variable bias and ensures that the estimated relationship between governance capacity and adaptation outcomes is not driven by alternative explanations. This strengthens the internal validity and credibility of the findings.

6. Result:

6.1 Descriptive Statistics: Overall Summary of Main Variables

Table 1 presents the descriptive statistics for all variables included in the analysis. The dataset consists of panel data covering 27 EU countries (Austria, Belgium, Bulgaria, Croatia, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Poland, Portugal, Romania, Slovakia, Slovenia, Spain, Sweden) across the years 1995 to 2023, allowing the study to examine both cross-country differences and within-country changes over time. The descriptive statistics provide an overview of the central tendencies and variation of the variables used in the empirical analysis.

The dependent variable, climate adaptation performance (*adapt_perf*), was constructed from the ND-GAIN vulnerability index and transformed into a 0–100 scale, where higher values indicate stronger adaptation performance and lower climate vulnerability. The average adaptation performance across the sample is 66.39, with a standard deviation of 3.41. The values range from 56.16 to 74.33, suggesting moderate but meaningful variation in adaptation performance across EU countries and over time. This variation indicates that while EU countries generally perform at relatively moderate to high adaptation levels, important differences still exist between countries and across years. The main independent variable, governance capacity (*wbgi_gee*), has a mean value of 1.06 and ranges from -0.88 to 2.32.⁷ This indicates substantial variation in government effectiveness across the countries included in the sample. The relatively high variation in governance capacity is particularly important for this study because it provides sufficient cross-country and temporal variation to examine whether stronger governance systems are associated with improved climate adaptation performance and reduced climate vulnerability. In other words, the variation in governance capacity allows the analysis to assess whether countries with more effective institutions and administrative systems tend to perform better in climate adaptation outcomes.

The control variables also display considerable variation across observations. Logged GDP per capita has a mean value of 9.94 and ranges from 7.22 to 11.81, reflecting substantial economic differences across EU member states. Logged population has a mean value of 15.80, while urban population

⁷ The WGI Government Effectiveness indicator is standardized globally (mean = 0; SD = 1) with a mean of zero and a standard deviation of one. Therefore, the average EU country in the sample scores approximately one standard deviation above the global mean in government effectiveness. (Kaufmann et al., 2010). Thus, the average EU country is just above 1 s.d. of the global mean.

share averages approximately 71.15 percent. Democracy levels are generally high within the sample, with a mean value of 0.84, although some variation remains across countries and years. These variations are important because economic development, demographic structure, and political institutions may also influence climate adaptation performance and therefore need to be controlled for in the regression analysis.

To further examine the panel structure of the dataset, within-country and between-country variation were assessed using the *xtsum* (Appendix, Table A1) procedure. The *xtsum* output separates the variation of each variable into two components: variation between countries and variation within countries over time. The results indicate that adaptation performance contains stronger between-country variation (3.41) than within-country variation (0.66), suggesting that long-term structural differences across countries explain a substantial proportion of the overall variation in adaptation performance. Similarly, governance capacity shows considerably larger between-country variation (0.66) compared to within-country variation (0.19). This indicates that governance capacity is relatively stable within countries over time, while larger differences exist across countries.

Overall, the descriptive statistics indicate that the dataset contains sufficient variation across the key variables and is suitable for panel-data analysis examining the relationship between governance capacity and climate adaptation performance.

Table 1: Summary Statistics: Main Variables

	Mean	SD	Min	Max	N
Adaptation Performance	66.394	3.413	56.162	74.325	783
Governance Capacity	1.056	0.677	-0.878	2.322	675
Log GDP per capita	9.942	0.865	7.216	11.813	783
Log Population	15.798	1.358	12.841	18.238	783
Urban Population Share	71.150	11.598	49.762	95.627	783
Democracy Level	0.837	0.085	0.390	0.923	783
Observations	783				

Note: The lower number of observations for governance capacity (N = 675) compared to the full sample (N = 783) is due to missing country-year observations in the Worldwide Governance Indicators dataset. In the earlier years of the study

period, the Government Effectiveness indicator was reported biannually rather than annually, resulting in systematic missing observations for alternating years (Kaufmann et al., 2010).

6.2.1 Descriptive: Governance Capacity and Adaptation Performance

Figure 1 presents the relationship between governance capacity and climate adaptation performance using country-level mean values across the observed period. The figure was constructed by averaging yearly observations for each country in order to isolate between-country variation and provide a clearer visual comparison among EU countries. Governance capacity is displayed on the horizontal axis, while climate adaptation performance is shown on the vertical axis. Each dot in the figure therefore represents the average governance capacity and average adaptation performance of a single country over the study period.

The scatterplot demonstrates a clear positive relationship between governance capacity and climate adaptation performance across the sample. Countries with higher average governance capacity generally tend to exhibit stronger adaptation performance, while countries with weaker governance systems tend to perform less effectively in climate adaptation outcomes. The fitted regression line further illustrates this positive trend across the countries included in the analysis.

Several country patterns are particularly visible in the figure. Countries such as Sweden, Denmark, Finland, and Netherlands appear in the upper-right section of the scatterplot, reflecting both high governance capacity and strong adaptation performance. In contrast, countries such as Romania and Bulgaria are positioned closer to the lower-left section of the figure, indicating comparatively lower governance effectiveness and weaker adaptation performance. These country-level differences suggest that stronger institutional effectiveness, higher administrative capacity, and more efficient public-sector coordination may contribute to improved adaptation outcomes and lower climate vulnerability (Adger et al., 2005; Biesbroek et al., 2015).

The correlation analysis further supports the visual relationship presented in Figure 1. Using country-level mean values, the Pearson correlation coefficient between governance capacity and climate adaptation performance is approximately 0.66 and statistically significant at the 1 percent level ($p < 0.001$), indicating a moderately strong positive association between the two variables across EU countries. Similarly, the Spearman rank correlation coefficient is 0.611 and also statistically significant ($p = 0.0009$), suggesting that countries ranking higher in governance effectiveness generally tend to achieve stronger climate adaptation performance. One notable case is Czechia,

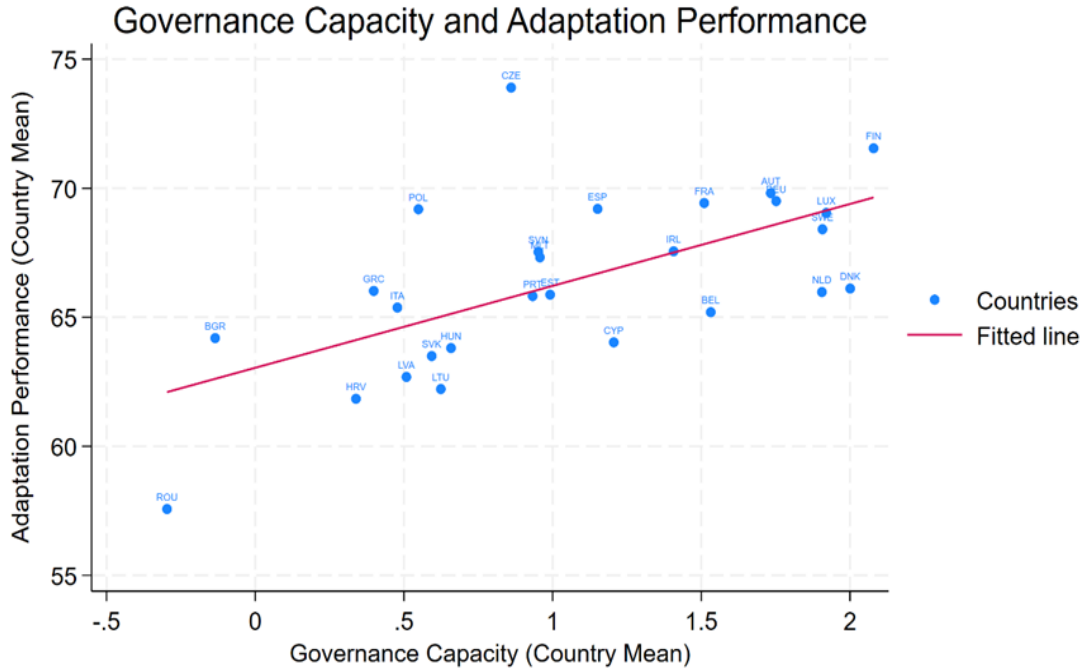
which appears as an outlier above the fitted regression line and consistently ranks among the strongest-performing countries in the dataset in terms of climate adaptation performance, indicating comparatively stronger adaptation outcomes relative to its level of governance capacity. Together, these findings provide preliminary empirical support for the theoretical expectation developed in this thesis that countries with stronger governance systems and more effective institutional capacity tend to perform better in implementing climate adaptation measures and reducing climate vulnerability (Fila et al., 2024; Biesbroek et al., 2015)

A boxplot (Appendix, Figure A1) provides comparing adaptation performance across high- and low-governance country groups. Countries were classified using the distribution of the governance capacity variable (*wbgi_gce*), where countries at or above the 75th percentile were categorized as high-governance countries and those at or below the 25th percentile were categorized as low-governance countries. The boxplot demonstrates that countries with stronger governance capacity generally exhibit higher median adaptation performance and lower variability compared to countries with weaker governance systems. Several outlier observations are also visible in the figure, indicating that some countries perform either substantially better or worse than the broader distribution within their governance category. Also, these outliers may reflect country-specific institutional, economic, or environmental conditions that are not fully captured by governance capacity alone.

By presenting time trends in climate adaptation performance across countries over the observed period (Appendix, Figure A2) further examines within-country changes over time. The figure illustrates that adaptation performance generally improved gradually over time across most EU countries, although the pace and consistency of improvement differ considerably between countries. Some countries display relatively stable trajectories, while others exhibit greater fluctuations across years. The figure also highlights the persistence of cross-country differences in adaptation performance throughout the study period, further supporting the conclusion that adaptation outcomes are shaped more strongly by long-term structural differences between countries than by short-term yearly variation within countries.

In summary, the descriptive analysis provides strong preliminary evidence of a positive relationship between governance capacity and climate adaptation performance across EU countries, while also indicating relatively “sticky” within-country variation over time. These findings establish an important empirical foundation for the multivariate panel analysis presented in the following sections.

Figure 1: Relationship Between Governance Capacity and Climate Adaptation Performance Across EU Countries (Country-Level Mean Values)



Note: Figure 1 displays country-level mean values averaged across the study period for 27 EU countries (N = 27). Country labels use standard ISO country codes (e.g., IRL = Ireland, SWE = Sweden, CZE = Czechia). Governance capacity is measured using the World Bank Government Effectiveness indicator, while adaptation performance is derived from the ND-GAIN vulnerability index transformed to a 0–100 scale. The fitted line represents the linear relationship between the two variables. No control variables are included in the figure

Importantly, the descriptive relationship illustrated in Figure 1 primarily reflects between-country or spatial variation rather than within-country temporal variation. The scatterplot was intentionally constructed using country-level mean values in order to highlight long-term structural differences across countries while minimizing yearly fluctuations over time. This distinction is particularly important given the panel structure of the dataset. The *xtsum* analysis (Appendix, Table A1) showed that both governance capacity and adaptation performance contain substantially larger between-country variation than within-country variation. Adaptation performance demonstrates between-country variation of 3.41 compared to within-country variation of only 0.66, while governance capacity exhibits between-country variation of 0.66 compared to within-country variation of 0.19. These findings suggest that both variables are relatively stable or sticky over time within individual

countries, while much larger structural differences exist across countries. This pattern is also consistent with the regression findings presented later in the analysis.

6.3. Overview of Baseline Models

Table 2 presents the baseline regression results examining the relationship between governance capacity and climate adaptation performance using four different model specifications: a cross-sectional model, pooled ordinary least squares (Pooled OLS), a fixed-effects (FE) baseline model, and a random-effects (RE) baseline model. These baseline models provide an initial assessment of the relationship before the inclusion of additional control variables in the main panel analysis. Together, the four specifications allow comparison between cross-country relationships and within-country changes over time using panel-data models specifically designed for repeated observations across countries and years.

The cross-sectional model, estimated using observations from the year 2020 only, shows a positive and statistically significant relationship between governance capacity and climate adaptation performance. The governance coefficient is 2.824 and statistically significant at the 1 percent level ($p < 0.01$), indicating that countries with stronger governance capacity tend to exhibit substantially higher adaptation performance. Substantively, this suggests that a 1 standard deviation increase in governance capacity is associated with an approximately 2.8-point increase in climate adaptation performance. The model explains approximately 30.2 percent of the variation in adaptation performance ($R^2 = 0.302$), indicating moderate explanatory power for a simple bivariate cross-sectional model. The constant terms are statistically significant across all models, reflecting the baseline level of adaptation performance when explanatory variables are held constant.

The pooled OLS model also shows a positive and statistically significant relationship between governance capacity and adaptation performance ($\beta = 2.936$, $p < 0.01$). However, this specification does not account for the panel structure of the data and assumes independence across all country-year observations. As a result, it may be affected by unobserved country-level heterogeneity. For this reason, the random-effects model is more appropriate as it explicitly accounts for the multi-level structure of the data by incorporating both within- and between-country variation. The RE results similarly show a positive and statistically significant association between governance capacity and adaptation performance.

The fixed-effects baseline model produces a smaller but still statistically significant coefficient for governance capacity ($\beta = 0.578$, $p < 0.05$). This reduction in the coefficient size is analytically important. The fixed-effects specification estimates whether changes in governance capacity within countries over time are associated with changes in climate adaptation performance. Unlike the cross-sectional and pooled OLS models, the FE model controls for stable country-specific characteristics that do not change substantially over time such as- geography, institutional history, administrative traditions, or long-standing economic structures. Consequently, the coefficient reflects only within-country variation over time rather than broader structural differences between countries. The smaller coefficient (0.578) in the fixed-effects model therefore suggests that the relationship between governance capacity and adaptation performance is weaker when focusing only on within-country changes over time. This is further explained by the fact that there is substantially less variation within countries than between countries in both governance capacity and adaptation performance, indicating that the relationship is primarily driven by cross-country structural differences rather than temporal changes within countries. Nevertheless, the coefficient remains statistically significant, providing evidence that improvements in governance capacity within countries are still positively associated with improved adaptation performance over time. The FE model explains approximately 41.3 percent of the variation in adaptation performance ($R^2 = 0.413$), indicating relatively strong explanatory power after accounting for country-level heterogeneity.

The random-effects baseline model also demonstrates a positive and statistically significant relationship between governance capacity and adaptation performance, with a coefficient of 0.622 ($p < 0.05$). The similarity between the FE and RE coefficients suggests a relatively stable relationship across alternative panel-data specifications. Unlike the FE model, the RE specification incorporates both within-country and between-country variation simultaneously while assuming that unobserved country-specific effects are not systematically correlated with the explanatory variables. Because the RE model retains cross-country information, the coefficient remains slightly larger than the FE estimate. The consistency of the positive governance coefficient across all four baseline specifications strengthens confidence in the overall empirical relationship between governance capacity and climate adaptation performance. The intraclass correlation coefficient ($ICC = 0.97$) indicates that the vast majority of variation in climate adaptation performance is explained by differences between countries, while only a very small share is attributable to within-country changes over time.

All together, the baseline models provide strong preliminary support for the study’s central theoretical expectation that stronger governance capacity is associated with improved climate adaptation performance across EU countries. At the same time, the reduction in coefficient magnitude between the cross-sectional/pooled models and the fixed-effects model suggests that the relationship is driven more strongly by structural differences between countries than by short-term within-country changes over time. This finding is also consistent with the earlier descriptive analysis and *xtsum* results, which showed that both governance capacity and adaptation performance contain substantially greater between-country variation than within-country variation

Table 2 Baseline Results: Governance Capacity and Adaptation Performance

	(1) Cross-section	(2) Pooled OLS	(3) FE Baseline	(4) RE Baseline
Governance Capacity	2.824*** (0.960)	2.936*** (0.775)	0.578** (0.280)	0.622** (0.283)
Constant	63.720*** (1.296)	63.065*** (0.963)	65.183*** (0.273)	65.144*** (0.758)
Observations	27	675	675	675
R^2	0.302	0.356	0.413	
Adjusted R^2	0.274	0.332	0.391	
ICC	—	—	—	0.972

Standard errors in parentheses

Year fixed effects included but omitted for readability., Standard errors clustered at country level where applicable.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

ICC reported from the Random Effects model and indicates the proportion of total variance attributable to between-country differences.

6.4 Main Panel Regression Analysis: Fixed and Random Effects Models

Table 3 presents the main panel regression results examining the relationship between governance capacity and climate adaptation performance using both fixed-effects (FE) and random-effects (RE) models. The specifications progressively incorporate additional controls, including log GDP per capita, log population, urban population share, democracy level, and year fixed effects, allowing for a more comprehensive assessment of the governance-adaptation relationship across EU countries over time.

Across the simpler specifications, governance capacity is positively and statistically significant. In the FE baseline model, the coefficient is positive and significant at the 5% level ($\beta = 0.578$, $SE = 0.280$, $p < 0.05$), while in the RE baseline model the effect is slightly larger and also significant ($\beta = 0.622$, $SE = 0.283$, $p < 0.05$). These results indicate that higher governance capacity is associated with higher levels of climate adaptation performance when only governance is considered.

However, once the full set of control variables is introduced, the effect of governance capacity becomes statistically insignificant in both FE and RE specifications. In the full fixed-effects model, the coefficient turns slightly negative and close to zero ($\beta = -0.017$, $SE = 0.314$, $p > 0.10$), while in the full random-effects model it remains positive but statistically insignificant ($\beta = 0.105$, $SE = 0.321$, $p > 0.10$).

Among the control variables, log GDP per capita shows a positive but statistically insignificant effect in the FE model ($\beta = 0.586$, $SE = 0.354$), while it becomes positive and statistically significant in the RE specification ($\beta = 0.735$, $SE = 0.326$, $p < 0.05$). Population size is negatively associated with adaptation performance in both models, although statistically insignificant (FE: $\beta = -1.791$, $SE = 1.113$; RE: $\beta = -0.744$, $SE = 0.581$). Urban population share and democracy level are also statistically insignificant across both specifications.

The fixed-effects specification explains a substantial proportion of within-country variation in climate adaptation performance, with an R^2 of approximately 0.482, compared to 0.413 in the baseline FE model. Across all specifications, the models are estimated using 675 observations, with standard errors clustered at the country level to account for heteroskedasticity and within-panel correlation. Year fixed effects are included in all models but omitted from the reported tables for clarity.

To determine the most appropriate panel-data specification, a Hausman specification test was conducted comparing the fixed-effects and random-effects estimators. The test result was statistically insignificant ($\chi^2 = 25.51$, $p = 0.6517$), indicating that the random-effects specification is statistically appropriate for the dataset. Nevertheless, the fixed-effects estimates remain substantively important because they specifically capture within-country changes over time.

The distinction between the FE and RE findings is further supported by the *xtsum* analysis and intraclass correlation coefficient (ICC) results. The ICC values from the random-effects models are extremely high, ranging approximately between 0.97 and 0.99 across specifications. This indicates

that most variation in climate adaptation performance is attributable to stable between-country differences rather than within-country yearly fluctuations. Given that only a very small proportion(only around 3%) of the total variation occurs within countries over time, it becomes more difficult for the fixed-effects models to identify stable and statistically significant and consistent relationships once additional controls are included. Consequently, Governance capacity demonstrates limited within-country variation over time, suggesting that institutional quality and administrative effectiveness evolve gradually rather than rapidly.

Table 3: Main Panel Regression Results: Fixed-Effects and Random-Effects Models of Governance Capacity and Climate Adaptation Performance

	(1) FE Baseline	(2) RE Baseline	(3) Fixed Effects	(4) Random Effects
Governance Capacity	0.578** (0.280)	0.622** (0.283)	-0.017 (0.314)	0.105 (0.321)
Log GDP per capita			0.586 (0.354)	0.735** (0.326)
Log Population			-1.791 (1.113)	-0.744 (0.581)
Urban Population Share			-0.014 (0.067)	-0.009 (0.062)
Democracy Level			-0.216 (0.961)	-0.329 (0.962)
Constant	65.183*** (0.273)	65.144*** (0.758)	89.661*** (18.692)	71.426*** (10.259)
Observations	675	675	675	675
R^2	0.413		0.482	
Adjusted R^2	0.391		0.459	

Standard errors in parentheses

Year fixed effects included but omitted for readability.,

Standard errors clustered at country level., ICC reported from Random Effects model (ICC = 0.972).

The analysis covers 27 EU countries over the period 1995-2023

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

6.5. Robustness check:

Several robustness checks were conducted to evaluate the stability and reliability of the estimated relationship between governance capacity and adaptation performance. These tests examine whether the main results are sensitive to alternative model specifications, lag structures, time periods, and sample composition.

First, a *reduced fixed-effects* specification was estimated including only governance capacity and democracy level. In this reduced model, governance capacity remains positive and statistically significant ($\beta = 0.605$, $SE = 0.292$, $p < 0.05$), indicating that the core relationship is not driven by the inclusion of additional control variables. However, the weakening of the governance coefficient in the fully specified models suggests that governance capacity and GDP per capita may partly capture overlapping structural characteristics within countries over time. Although the VIF diagnostics do not indicate problematic multicollinearity, the two variables may reflect related dimensions of long-term socioeconomic and institutional development. Democracy level remains statistically insignificant ($\beta = -0.700$, $SE = 1.071$), suggesting limited explanatory power in this simplified specification. The model explains 41.6% of the variation in adaptation performance ($R^2 = 0.416$), providing a baseline confirmation of the governance-adaptation relationship.

Second, a *lagged governance* specification was estimated to address potential reverse causality⁸ and to capture delayed effects of institutional quality on adaptation outcomes. When governance capacity is lagged by one period, the coefficient becomes small and statistically insignificant ($\beta = -0.018$, $SE = 0.322$), suggesting that short-term lagged effects are weak or not immediately observable. This implies that governance capacity may influence adaptation performance through longer-term institutional channels rather than immediate yearly changes. The model fit improves slightly

⁸ Reverse causality refers to a situation where the direction of influence between two variables is unclear, meaning that while governance capacity may affect climate adaptation performance, it is also possible that better adaptation outcomes contribute to improvements in governance capacity over time (Wooldridge, 2010).

compared to the reduced model ($R^2 = 0.467$), but the governance effect loses statistical significance in this specification.

Third, the analysis was restricted to the *post-2010 period* ($N = 378$) to examine whether the relationship holds under more recent conditions characterized by intensified EU climate governance and adaptation policy coordination. In this subsample, governance capacity becomes statistically insignificant ($\beta = -0.071$, $SE = 0.369$), while key structural variables remain relevant: population size retains a significant negative effect ($\beta = -2.165$, $p < 0.05$), and GDP per capita remains positive but imprecisely estimated ($\beta = 0.604$, $SE = 0.377$). The model explains a more modest share of variation in this restricted period ($R^2 = 0.227$), indicating weaker explanatory power in the post-2010 subsample.

Finally, a sample restriction *excluding high-income* countries was applied to test whether the results are driven primarily by wealthier EU member states. In this specification ($N = 483$), governance capacity remains positive but statistically insignificant ($\beta = 0.032$, $SE = 0.208$), suggesting that the effect is not robust when richer countries are excluded. However, population size remains negatively associated with adaptation performance ($\beta = -2.551$, $p < 0.10$), while urban population share becomes weakly positive and marginally significant ($\beta = 0.087$, $p < 0.10$). The explanatory power of the model is relatively high in this subsample ($R^2 = 0.645$), indicating substantial variation explained by structural factors even when governance effects weaken.

In summary, the robustness checks provide a mixed but informative pattern. While the positive association between governance capacity and adaptation performance is stable in the baseline and reduced specifications, its magnitude and statistical significance decline in lagged, time-restricted, and income-restricted models. At the same time, structural variables, particularly population size, show relatively more stable effects across specifications. Thus, these results suggest that governance capacity is an important but context-sensitive determinant of adaptation performance, with its influence varying across time, development levels, and model specifications.

Table 4: Robustness Checks: Governance and Adaptation Performance

(1)	(2)	(3)	(4)
Reduced Model	Lagged Governance	Post-2010 Sample	Excluding Rich Countries

Governance Capacity	0.605**		-0.071	0.032
	(0.292)		(0.369)	(0.208)
Democracy Level	-0.700	-0.135	0.227	0.551
	(1.071)	(0.915)	(0.824)	(0.699)
L.Governance Capacity		-0.018		
		(0.322)		
Log GDP per capita		0.566	0.604	0.368
		(0.371)	(0.377)	(0.416)
Log Population		-1.807	-2.165**	-2.551*
		(1.059)	(0.979)	(1.251)
Urban Population Share		-0.009	-0.036	0.087*
		(0.065)	(0.103)	(0.043)
Constant	65.736***	89.769***	96.964***	95.615***
	(0.870)	(18.063)	(14.464)	(21.483)
Observations	675	648	378	483
R^2	0.416	0.467	0.227	0.645

Standard errors in parentheses

All models include year fixed effects., Clustered standard errors at country level.,

Robustness checks examine specification sensitivity, temporal stability, and sample dependence.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

6.6 Regression Diagnostics:

Several diagnostic tests were conducted to evaluate the validity of the panel regression models and assess the robustness of the empirical strategy.

First, multicollinearity among the explanatory variables was examined using the Variance Inflation Factor (VIF). The results indicate no evidence of problematic multicollinearity, as all VIF values

remain well below conventional threshold levels. ⁹The highest VIF is observed for governance capacity (VIF = 3.52), followed by GDP per capita (VIF = 2.88). Although these values remain below conventional thresholds for multicollinearity, they indicate a moderate correlation between governance capacity and GDP per capita. This overlap may partly explain why the statistical significance of governance capacity weakens in the fully specified models, as both variables capture related aspects of broader structural development, while democracy level (VIF = 1.60), urban population share (VIF = 1.47), and population size (VIF = 1.02) demonstrate comparatively low values. The mean VIF of 2.10 further suggests that the explanatory variables are not excessively correlated with one another and therefore do not create serious instability in the coefficient estimates.

To improve the reliability of statistical inference under these panel-data conditions, the main fixed-effects models were estimated using country-level clustered standard errors based on 27 country clusters. Clustered standard errors help account for potential heteroskedasticity and within-panel correlation over time, thereby providing more reliable and conservative standard error estimates.

Lastly, an outlier¹⁰ sensitivity analysis was conducted to examine whether the results were disproportionately influenced by extreme observations. This was implemented by trimming the top 1% of values of the dependent variable, reducing the sample size from 675 to 668 observations. The re-estimated fixed-effects model produced substantively similar results. Governance capacity remained statistically insignificant ($\beta = 0.003$, $p = 0.993$), GDP per capita remained positively associated with adaptation performance though statistically weaker ($\beta = 0.578$, $p = 0.116$), and population size retained a negative relationship ($\beta = -1.759$, $p = 0.126$). Importantly, the overall pattern and direction of the coefficients remained largely stable, suggesting that the main findings are not driven by a small number of extreme observations.

Thereby, the diagnostic tests show that the data have some common panel-data issues, but the estimation methods used are still appropriate and the results remain reliable. The use of fixed

⁹ VIF values above 10 are commonly interpreted as indicating severe multicollinearity, although lower thresholds are sometimes used depending on model context (Gujarati & Porter, 2009).

¹⁰ Outlier sensitivity tests assess whether regression results are disproportionately influenced by extreme observations or unusually large values within the dataset (Osborne & Overbay, 2004).

effects, clustered standard errors, and outlier sensitivity analysis increases confidence in the robustness and reliability of the empirical findings.

7. Discussion

In relation to the main research question whether stronger governance capacity reduces the climate adaptation implementation gap within and across EU countries the findings provide only partial and conditional support. Consistent with Hypothesis 1 and the first sub-research question, the baseline models show a positive and statistically significant association between governance capacity and climate adaptation performance indicating that countries with stronger governance tend to achieve better adaptation outcomes in simpler specifications. However, this relationship becomes statistically insignificant once key socioeconomic controls such as GDP per capita, population size, urbanization, and democracy are introduced. This suggests that the observed governance-adaptation association is not robust to the inclusion of broader structural factors and is likely shaped by underlying economic and demographic conditions. Therefore, the results indicate that governance capacity is associated with adaptation performance but does not independently explain variation across countries when structural differences are taken into account. These findings are consistent with prior research emphasizing the role of structural economic conditions in shaping adaptation capacity (Adger et al., 2005; Biesbroek et al., 2015).

In relation to the sub-research questions and Hypothesis 2, the results further suggest that within-country improvements in governance capacity are not strongly associated with short-term improvements in adaptation performance, as reflected in the statistically insignificant fixed-effects estimates in the full specification. So, this indicates that governance capacity may influence adaptation outcomes indirectly through its interaction with broader structural conditions such as economic development and population structure, rather than exercising a strong standalone effect. This interpretation is consistent with prior research emphasizing that climate adaptation capacity is shaped by broader socioeconomic and institutional contexts rather than governance quality alone (e.g., Adger et al., 2005; Biesbroek et al., 2015; Dupuis & Biesbroek, 2013).

Finally, the pattern across fixed and random effects models suggests that much of the variation in adaptation performance is driven by between-country structural differences rather than within-country changes over time. This is consistent with the relatively stable or “sticky” nature of the main

variables, which display limited within-country variation over time, leaving less variation for the fixed-effects model to capture once controls are included. In summary, the findings provide only partial support for the hypotheses and suggest that the governance-adaptation relationship is conditional and context-dependent rather than uniformly direct or causal. Thus, the results provide a consistent but context-dependent picture of the governance-adaptation relationship.

In the baseline specifications, governance capacity is positively and statistically significantly associated with climate adaptation performance across all models. However, this relationship weakens substantially once socioeconomic controls are introduced in the full panel models. Consequently, this outline suggests that governance capacity is correlated with adaptation outcomes primarily because it co-varies with broader structural conditions, particularly economic development and population size. In other words, governance capacity mainly reflects a country's overall level of development and institutional strength, rather than acting as a separate cause entirely separate driver of better adaptation performance once those factors are included in the analysis. This finding is important because it questions a simplified interpretation often found in the climate governance literature, where governance quality is assumed to exert a direct and stable effect on adaptation outcomes. At the same time, the weaker within-country effects may also suggest that both governance capacity and adaptation performance are influenced by additional underlying structural factors over time, or that greater temporal and cross-country variation would be required to more fully identify these relationships.

A key contribution of this study lies in distinguishing between between-country (spatial) variation and within-country (temporal) variation. The combined evidence from FE/RE models, xtsum results, and ICC values shows that the majority of variation in adaptation performance is explained by stable differences across countries rather than changes within countries over time. This has two important implications. *First*, climate adaptation performance within EU countries is largely structurally “sticky”, meaning it changes slowly over time. Countries tend to maintain relatively stable positions in the distribution of adaptation performance. *Second*, governance capacity itself also shows limited within-country variation, suggesting that institutional quality evolves gradually rather than fluctuating strongly year to year. This reinforces the idea that both governance and adaptation are deeply embedded in long-term institutional and developmental trajectories. As a result, the governance-adaptation relationship is primarily spatial rather than temporal, meaning that cross-

country differences matter more than short-term policy or institutional changes within countries. In this sense, governance capacity should be understood less as a short-term policy instrument and more as a broader institutional characteristic embedded within wider development processes.

Control variables support this interpretation. GDP per capita shows a stable positive association with adaptation performance, suggesting that economic resources and long-term development strongly enable effective adaptation. Population size is consistently negatively associated with adaptation outcomes, indicating that larger administrative systems face greater coordination and implementation challenges. In contrast, urbanization and democracy show weak or non-robust effects, implying their influence is likely indirect through broader structural conditions. Overall, these findings suggest that the adaptation implementation gap across EU countries is driven not only by governance quality but also by deeper structural inequalities and path-dependent differences in economic and institutional development.

Finally, the robustness and diagnostic analyses strengthen the credibility of the findings. Governance effects weaken across lagged and fully controlled specifications, while stable coefficient patterns, low multicollinearity, clustered standard errors, and robustness checks all support the reliability of the empirical results. Also, this result appears in all the different models and tests used in the study, so it is a key and reliable finding of the research. Consequently, this finding is consistent across multiple empirical approaches and represents one of the most important contributions of the study.

8. Limitations of the Data Materials and Methodology

Although this study uses internationally recognized datasets and a structured quantitative research design, several limitations should be acknowledged. These limitations do not undermine the overall analysis but help clarify the scope and interpretation of the findings.

First, some of the variables used in the study are composite or proxy indicators. For example, Government Effectiveness, the Electoral Democracy Index, and the ND-GAIN vulnerability index combine multiple dimensions into single numerical measures. While this improves cross-national comparability and makes large-scale analysis possible, such indicators may not fully capture all country-specific or sector-specific aspects of governance and climate adaptation (Fukuyama, 2013; Cingolani, 2013). Nevertheless, these measures are widely used in comparative governance and

climate research and provide consistent data across countries and years (Teorell et al., 2026; Chen et al., 2015).

Second, climate adaptation performance is inherently difficult to measure because adaptation is a complex and multidimensional process. Although the ND-GAIN vulnerability index provides a broad measure of climate-related vulnerability it may not fully capture local adaptation practices, implementation quality, or country-specific differences within sectors and regions (Chen et al., 2015; Dupuis & Biesbroek, 2013). However, the index remains one of the most internationally comparable and widely used measures of climate vulnerability and adaptation-related performance, making it appropriate for comparative panel analysis (ND-GAIN, 2023).

Third, some variables in the study rely partly on expert assessments and aggregated indicators, particularly the Electoral Democracy Index and governance-related measures. As a result, a certain degree of measurement uncertainty or subjective interpretation may exist. However, these datasets are produced by internationally recognized institutions and are widely used in comparative political and public policy research (Teorell et al., 2026).

Fourth, the study focuses only on EU member states. This improves comparability because EU countries share relatively similar institutional and economic contexts. However, it also means that the findings may not be fully generalizable to non-European or lower-income countries where governance systems and climate vulnerabilities may differ significantly. Some missing observations were present in the Government Effectiveness variable(675), which reduced the final number of usable observations. However, the dataset still retains a sufficiently large and balanced panel for analysis.

Finally, although panel data analysis improves the ability to examine relationships across countries and over time, the study primarily identifies statistical associations and establishing fully definitive causal relationships remains difficult because climate adaptation outcomes are influenced by many interconnected political, economic, environmental, and social factors that cannot be fully captured within a single empirical model (IPCC, 2014). Despite these limitations, the study uses high-quality and internationally recognized datasets, including ND-GAIN, QoG, World Bank, and V-Dem. The combination of longitudinal panel data, theoretically informed variable selection, and multiple control variables provide a reliable and transparent framework for examining the relationship between governance capacity and climate adaptation performance across and within EU countries.

9. Conclusion

This thesis examined how governance capacity influences climate adaptation performance across EU countries and whether this relationship operates differently across countries and over time. Three main findings emerge. First, there is a clear positive association between governance capacity and climate adaptation performance in baseline models, suggesting that countries with stronger institutions tend to achieve better adaptation outcomes. Second, this relationship weakens substantially once structural economic and demographic factors are included, indicating that governance does not operate independently but rather alongside broader development conditions. Third, adaptation performance is driven primarily by between-country structural differences rather than within-country changes over time, as shown by fixed-effects results, xtsum decomposition, and ICC estimates. In general, the findings suggest that governance capacity shapes climate adaptation performance, but not in a uniform or standalone manner. Its influence is largely indirect and conditioned by structural factors such as economic development and long-term institutional capacity. Moreover, governance explains differences in adaptation performance more strongly across countries than within countries over time, indicating that adaptation capacity is deeply rooted in structural national characteristics rather than short-term institutional improvements.

These results contribute to the climate governance and adaptation literature by questioning the assumption of a direct governance-performance link. Instead, they support a more structural and path-dependent understanding of adaptation capacity, where governance interacts with economic and demographic conditions rather than operating independently. Consequently, from a policy perspective, the findings suggest that improving climate adaptation performance requires more than strengthening governance indicators alone. While institutional quality remains important, effective adaptation policy must also address economic constraints, administrative capacity inequalities, demographic pressures, long-term structural development gaps. This implies that to reduce climate adaptation implementation gap EU climate adaptation policy should combine governance reforms with broader development-oriented support, particularly for lower-performing countries. In sum, climate adaptation performance in EU countries should be understood not as a direct result of governance quality alone, but as a structurally embedded, path-dependent process shaped by interacting dimensions of economic development and institutional capacity. To conclude, these factors influence how effectively countries can translate adaptation policies into practice and, ultimately, help explain persistent differences in the climate adaptation implementation gap.

10. References

- Adger, W. N., Arnell, N. W., & Tompkins, E. L. (2005). Successful adaptation to climate change across scales. *Global Environmental Change*, 15(2), 77–86.
<https://doi.org/10.1016/j.gloenvcha.2004.12.005>
- Bachtler, J., Mendez, C., & Oraže, H. (2014). From conditionality to Europeanization in Central and Eastern Europe: Administrative performance and capacity in cohesion policy. *European Planning Studies*, 22(4), 735–757. <https://doi.org/10.1080/09654313.2013.772744>
- Baltagi, B. H. (2021). *Econometric analysis of panel data* (6th ed.). Springer.
- Bachtrögl, J. (2024). The role of administrative capacity for an effective implementation of EU cohesion policy (ZEW Discussion Paper No. 24-067). ZEW – Leibniz Centre for European Economic Research.
- Biesbroek, R., Klostermann, J., Termeer, C., & Kabat, P. (2015). On the nature of barriers to climate change adaptation. *Regional Environmental Change*, 13(5), 1119–1129.
<https://doi.org/10.1007/s10113-013-0421-y>
- Cingolani, L. (2013). The state of state capacity: A review of concepts, evidence and measures (UNU-MERIT Working Paper No. 2013-053). UNU-MERIT.
- Coppedge, M., Gerring, J., Knutsen, C. H., Lindberg, S. I., Teorell, J., Altman, D., Bernhard, M., Fish, M. S., Glynn, A., Hicken, A., Lührmann, A., Marquardt, K. L., McMann, K., Paxton, P., Pemstein, D., Seim, B., Sigman, R., Skaaning, S.-E., Staton, J., ... Ziblatt, D. (2024). V-Dem Codebook v14. Varieties of Democracy (V-Dem) Project, University of Gothenburg.
https://www.vdem.net/documents/44/vdem_codebook_v14.pdf
- Dupuis, J., & Biesbroek, R. (2013). Comparing apples and oranges: The dependent variable problem in comparing and evaluating climate change adaptation policies. *Global Environmental Change*, 23(6), 1476–1487. <https://doi.org/10.1016/j.gloenvcha.2013.07.022>
- Eakin, H., et al. (2018). The governance of climate adaptation: Institutional complexity and the challenge of scale. *Current Opinion in Environmental Sustainability*, 12, 1–9.
- European Environment Agency. (2021). *Urban adaptation in Europe: How cities and towns respond to climate change*. European Environment Agency.
- European Environment Agency. (2025). *From adaptation planning to action: Insights into progress and challenges across Europe*. European Environment Agency.

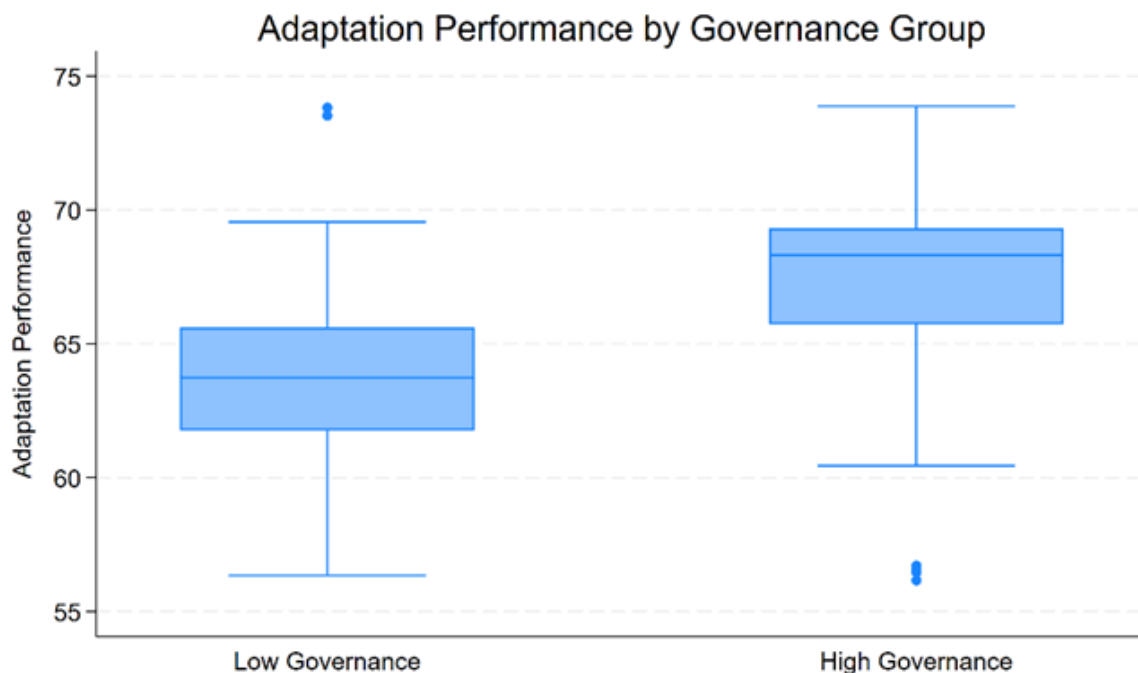
- Fila, D., Fünfgeld, H., & Dahlmann, H. (2024). Climate change adaptation with limited resources: Adaptive capacity and action in small- and medium-sized municipalities. *Environment, Development and Sustainability*, 26(3), 5607–5627. <https://doi.org/10.1007/s10668-023-02999-3>
- Fukuyama, F. (2013). What is governance? *Governance*, 26(3), 347–368. <https://doi.org/10.1111/gove.12035>
- Fukuyama, F. (2016). Governance: What do we know, and how do we know it? *Annual Review of Political Science*, 19(1), 89–105. <https://doi.org/10.1146/annurev-polisci-042214-044240>
- Gordon, D. I. (2009). Constructing a bid protest process: Choices every procurement challenge system must make. *Public Contract Law Journal*, 35(3), 427–458.
- Görbig, G. M., Flacke, J., Treville, A., Sliuzas, R., & Reckien, D. (2026). Empirical insights on the adaptation planning-implementation gap from the Global Covenant of Mayors European signatories. *npj Urban Sustainability*, 6, 66. <https://doi.org/10.1038/s42949-026-00365-6>
- Grindle, M. S. (1996). *Challenging the state: Crisis and innovation in Latin America and Africa*. Cambridge University Press.
- Gerring, J., & Christenson, D. (2017). *Applied social science methodology: An introductory guide*. Cambridge University Press.
- Incaltarau, C., Pascariu, G. C., & Surubaru, N. C. (2020). Evaluating the determinants of EU funds absorption across old and new member states: The role of administrative capacity and political governance. *Journal of Common Market Studies*, 58(4), 941–961. <https://doi.org/10.1111/jcms.12995>
- Intergovernmental Panel on Climate Change. (2014). *Climate change 2014: Impacts, adaptation, and vulnerability*. Cambridge University Press.
- Jonas Meckling, & Jonas Nahm. (2018). The power of process: State capacity and climate policy. *Governance*, 31(4), 741–757. <https://doi.org/10.1111/gove.12338>
- Kaufmann, D., Kraay, A., & Mastruzzi, M. (2010). *The Worldwide Governance Indicators: Methodology and analytical issues*. World Bank Policy Research Working Paper No. 5430. World Bank
- Klein, R. J. T., et al. (2014). Adaptation opportunities, constraints, and limits. In *Climate change 2014: Impacts, adaptation, and vulnerability*. Cambridge University Press.
- Kellstedt, P. M., & Whitten, G. D. (2018). *The fundamentals of political science research* (3rd ed.). Cambridge University Press.

- Magnan, A. K., Schipper, E. L. F., Burkett, M., Bharwani, S., Burton, I., Eriksen, S., Gemenne, F., Schaar, J., & Ziervogel, G. (2016). Addressing the risk of maladaptation to climate change. *Wiley Interdisciplinary Reviews: Climate Change*, 7(5), 646–665.
<https://doi.org/10.1002/wcc.409>
- Mendez, C., & Bachtler, J. (2022). The quality of government and administrative performance: Explaining cohesion policy compliance, absorption and achievements across EU regions. *Regional Studies*. <https://doi.org/10.1080/00343404.2021.1965346>
- Moser, S. C., & Ekstrom, J. A. (2010). A framework to diagnose barriers to climate change adaptation. *Proceedings of the National Academy of Sciences*, 107(51), 22026–22031.
<https://doi.org/10.1073/pnas.1007887107>
- Moyo, T. N., C. P., & Mashingaidze, T. M. (2026). Multi-level governance and climate change adaptation: A systematic literature review. *Sustainable Futures*, 11, 101724.
<https://doi.org/10.1016/j.sftr.2026.101724>
- Notre Dame Global Adaptation Initiative. (2024). ND-GAIN Country Index [Data set]. University of Notre Dame. <https://gain.nd.edu/our-work/country-index/>
- Povitkina, M. (2018). The limits of democracy in tackling climate change. *Environmental Politics*, 27(3), 411–432. <https://doi.org/10.1080/09644016.2018.1444723>
- Pressman, J. L., & Wildavsky, A. (1973). *Implementation: How great expectations in Washington are dashed in Oakland*. University of California Press.
- Roest, A. H., Restemeyer, B., van den Brink, M., Horlings, I., & Boogaard, F. C. (2025). Unravelling the policy-to-implementation gap: Climate adaptation governance arrangements in medium-sized cities. *Environmental Policy and Governance*.
<https://doi.org/10.1002/eet.70042>
- Rothstein, B., & Teorell, J. (2008). What is quality of government? A theory of impartial government institutions. *Governance*, 21(2), 165–190. <https://doi.org/10.1111/j.1468-0491.2008.00391.x>
- Savoia, A., & Sen, K. (2015). Measurement, evolution, determinants, and consequences of state capacity: A review of recent research. *Journal of Economic Surveys*, 29(3), 441–458.
<https://doi.org/10.1111/joes.12101>
- Sundström, A., Harring, N., Jagers, S. C., & Povitkina, M. (2026). The impact of corruption on climate change mitigation: A review article. *Environmental Politics*, 1–25.
- Teorell, J., Sundström, A., Holmberg, S., Rothstein, B., Alvarado Pachon, N., Phiri, V. S., Chen, C., & Liu, Z. (2026). The Quality of Government Standard Dataset (Version Jan26) [Data set]. The Quality of Government Institute, University of Gothenburg.
<https://doi.org/10.18157/qogstdjan26>

- Terracciano, B., & Graziano, P. R. (2016). EU cohesion policy implementation and administrative capacities: Insights from Italian regions. *Regional & Federal Studies*, 26(3), 293–320. <https://doi.org/10.1080/13597566.2016.1162645>
- Tosun, J. (2014). Absorption of regional funds: A comparative analysis. *Journal of Common Market Studies*, 52(2), 371–387. <https://doi.org/10.1111/jcms.12088>
- United Nations Environment Programme. (2021). *Adaptation gap report 2021: The gathering storm*. United Nations Environment Programme.
- Wooldridge, J. M. (2013). *Introductory econometrics: A modern approach* (5th ed.). South-Western Cengage Learning.

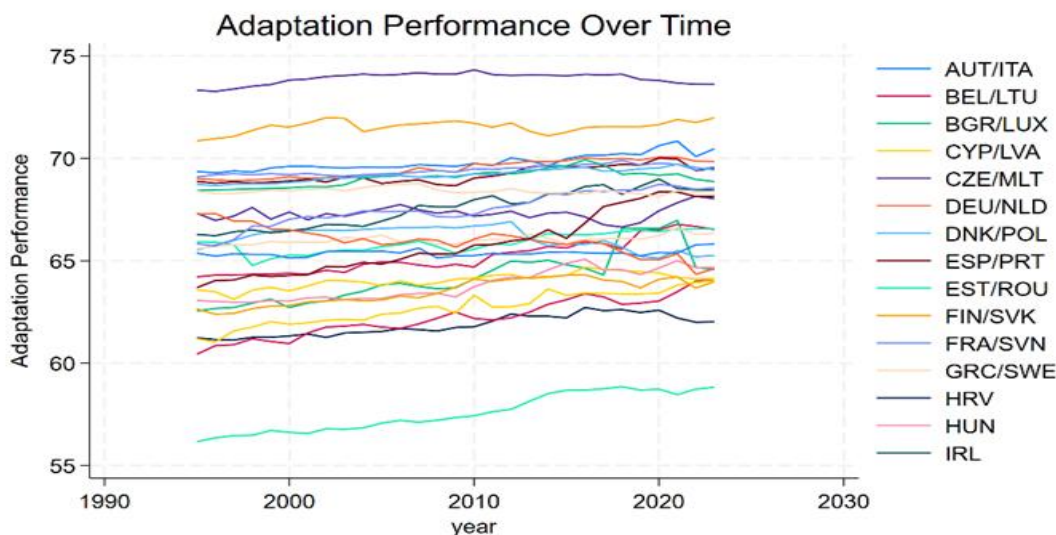
Appendix:

Figure A1: Distribution of Adaptation Performance Across Governance Groups



Note: Countries were grouped using the 25th and 75th percentiles of the governance capacity variable (wbgi_gee). The figure compares the distribution of climate adaptation performance between high- and low-governance country groups.

Figure A2: Country-Level Time Trends in Climate Adaptation Performance



Note: The figure displays yearly trends in climate adaptation performance across EU countries over the observed period. The visualization illustrates both within-country temporal changes and persistent cross-country differences over time

Table A1: Panel Descriptive Statistics (xtsum Results): Within and Between Variation

Variable	Overall Mean	Overall SD	Min	Max	Between SD	Within SD	N	Countries (n)	Time (T)
Adaptation Performance	66.39	3.41	56.16	74.33	3.41	0.66	783	27	29
Governance Capacity	1.06	0.68	-0.88	2.32	0.66	0.19	675	27	25
Log GDP per capita	9.94	0.87	7.22	11.81	0.74	0.47	783	27	29
Log Population	15.80	1.36	12.84	18.24	1.38	0.07	783	27	29
Urban Population Share	71.15	11.60	49.76	95.63	11.60	2.17	783	27	29
Democracy (V-Dem Polyarchy)	0.84	0.08	0.39	0.92	0.07	0.05	783		

Note: The table is based on the xtsum command and reports overall, between-country, and within-country variation for the main variables in the panel dataset. The statistics are included to assess how much variation occurs across countries versus within countries over time, which is important for interpreting the fixed-effects and random-effects models.

Table A2: Variance Inflation Factor (VIF).

Variable	VIF	1/VIF
Governance Capacity	3.52	0.284
Logged GDP per capita	2.88	0.347
Democracy Label	1.60	0.624
Urban Population	1.47	0.679
Logged Population	1.02	0.980
Mean VIF	2.10	

Note: Variance Inflation Factor (VIF) statistics were calculated to assess potential multicollinearity among the explanatory variables.

Table A3: Comparison Table based on actual FE (baseline) and trimmed FE (outlier sensitivity) results:

Variable	Baseline FE (N=675)	Trimmed FE (N=668)
Governance	-0.0174 (0.124)	0.00259 (0.313)
Log GDP	0.586 (0.123)	0.578 (0.355)
Log Population	-1.791 (0.373)	-1.759 (1.111)
Urban Pop. Share	-0.0139 (0.012)	-0.0157 (0.067)
Democracy	-0.216 (0.404)	-0.265 (0.960)
Constant	89.661***	89.280***
Observations	675	668
R ² (within)	0.4818	0.4857

Note: Table-9 compares the baseline fixed-effects model with the outlier-trimmed specification excluding the top 1% of the dependent variable distribution. The results remain substantively consistent across both models.

Table A4: Full Regression results including year effects

	(1) FE Baseline	(2) RE Baseline	(3) Fixed Effects	(4) Random Effects
Governance Capacity	0.578** (0.280)	0.622** (0.283)	-0.017 (0.314)	0.105 (0.321)
year=1996	0.000 (.)	0.000 (.)	0.000 (.)	0.000 (.)
year=1998	0.167** (0.074)	0.168** (0.074)	0.145* (0.076)	0.139* (0.076)
year=2000	0.155* (0.090)	0.151* (0.091)	0.252*** (0.073)	0.235*** (0.075)
year=2002	0.239** (0.113)	0.229** (0.114)	0.328*** (0.080)	0.269*** (0.073)
year=2003	0.284** (0.114)	0.276** (0.114)	0.226* (0.124)	0.136 (0.106)
year=2004	0.370*** (0.110)	0.365*** (0.111)	0.192 (0.193)	0.080 (0.172)
year=2005	0.395*** (0.122)	0.387*** (0.123)	0.204 (0.204)	0.072 (0.182)
year=2006	0.498*** (0.115)	0.493*** (0.115)	0.226 (0.253)	0.082 (0.229)
year=2007	0.576*** (0.112)	0.572*** (0.112)	0.187 (0.326)	0.016 (0.302)
year=2008	0.496*** (0.118)	0.491*** (0.118)	0.063 (0.367)	-0.134 (0.335)

year=2009	0.478*** (0.135)	0.471*** (0.134)	0.143 (0.319)	-0.043 (0.288)
year=2010	0.698*** (0.141)	0.694*** (0.141)	0.338 (0.336)	0.160 (0.308)
year=2011	0.751*** (0.140)	0.746*** (0.141)	0.367 (0.360)	0.170 (0.329)
year=2012	0.825*** (0.148)	0.818*** (0.148)	0.506 (0.341)	0.312 (0.305)
year=2013	0.809*** (0.165)	0.801*** (0.165)	0.480 (0.360)	0.274 (0.321)
year=2014	0.926*** (0.191)	0.917*** (0.190)	0.596 (0.384)	0.381 (0.343)
year=2015	0.943*** (0.184)	0.935*** (0.183)	0.683* (0.341)	0.487 (0.302)
year=2016	1.092*** (0.198)	1.085*** (0.197)	0.818** (0.374)	0.614* (0.338)
year=2017	1.044*** (0.204)	1.037*** (0.203)	0.732* (0.423)	0.511 (0.385)
year=2018	1.052*** (0.237)	1.043*** (0.236)	0.709 (0.448)	0.466 (0.392)
year=2019	1.031*** (0.265)	1.022*** (0.265)	0.697 (0.459)	0.451 (0.407)
year=2020	1.130*** (0.282)	1.119*** (0.282)	0.838* (0.466)	0.583 (0.414)
year=2021	1.223*** (0.276)	1.210*** (0.276)	0.867 (0.513)	0.587 (0.452)
year=2022	0.994***	0.983***	0.641	0.361

	(0.274)	(0.273)	(0.518)	(0.464)
year=2023	1.081*** (0.260)	1.072*** (0.259)	0.645 (0.568)	0.348 (0.511)
Log GDP per capita			0.586 (0.354)	0.735** (0.326)
Log Population			-1.791 (1.113)	-0.744 (0.581)
Urban Population Share			-0.014 (0.067)	-0.009 (0.062)
Democracy Level			-0.216 (0.961)	-0.329 (0.962)
Constant	65.183*** (0.273)	65.144*** (0.758)	89.661*** (18.692)	71.426*** (10.259)
Observations	675	675	675	675
R^2	0.413		0.482	
Adjusted R^2	0.391		0.459	

Standard errors in parentheses

All models include year fixed effects shown explicitly.

,
Standard errors clustered at country level where applicable.

*
 $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A5: Full Robustness Models Including Year Effects

	(1) Reduced Model	(2) Lagged Governance	(3) Post-2010 Sample	(4) Excluding Rich Countries
Governance Capacity	0.605** (0.292)		-0.071 (0.369)	0.032 (0.208)
Democracy Level	-0.700 (1.071)	-0.135 (0.915)	0.227 (0.824)	0.551 (0.699)

year=1996	0.000 (.)			0.000 (.)
year=1998	0.170** (0.073)			0.119 (0.078)
year=2000	0.167* (0.089)			0.169* (0.095)
year=2002	0.248** (0.112)			0.239** (0.113)
year=2003	0.294** (0.113)	0.147 (0.132)		0.164 (0.159)
year=2004	0.381*** (0.108)	0.116 (0.203)		0.188 (0.220)
year=2005	0.406*** (0.120)	0.126 (0.227)		0.195 (0.245)
year=2006	0.511*** (0.112)	0.151 (0.266)		0.228 (0.311)
year=2007	0.592*** (0.110)	0.114 (0.346)		0.171 (0.405)
year=2008	0.514*** (0.116)	-0.009 (0.399)		0.026 (0.474)
year=2009	0.498*** (0.132)	0.067 (0.348)		0.106 (0.412)
year=2010	0.720*** (0.142)	0.263 (0.350)	0.000 (.)	0.300 (0.410)
year=2011	0.773*** (0.143)	0.291 (0.392)	0.033 (0.067)	0.330 (0.438)
year=2012	0.844***	0.428	0.178**	0.462

	(0.151)	(0.370)	(0.075)	(0.416)
year=2013	0.824*** (0.163)	0.403 (0.388)	0.160 (0.102)	0.518 (0.423)
year=2014	0.938*** (0.188)	0.519 (0.412)	0.282** (0.129)	0.697 (0.463)
year=2015	0.955*** (0.182)	0.603 (0.359)	0.376*** (0.106)	0.695* (0.404)
year=2016	1.097*** (0.197)	0.739* (0.397)	0.518*** (0.138)	0.816* (0.436)
year=2017	1.042*** (0.204)	0.654 (0.448)	0.440** (0.187)	0.713 (0.493)
year=2018	1.045*** (0.237)	0.632 (0.481)	0.424* (0.237)	0.820 (0.540)
year=2019	1.025*** (0.267)	0.620 (0.487)	0.417 (0.265)	0.774 (0.559)
year=2020	1.117*** (0.287)	0.760 (0.499)	0.566** (0.274)	0.932 (0.556)
year=2021	1.206*** (0.279)	0.791 (0.541)	0.599* (0.327)	0.976 (0.625)
year=2022	0.977*** (0.281)	0.566 (0.540)	0.378 (0.322)	0.780 (0.643)
year=2023	1.060*** (0.269)	0.573 (0.587)	0.385 (0.368)	0.740 (0.689)
L.Governance Capacity		-0.018 (0.322)		
Log GDP per capita		0.566	0.604	0.368

		(0.371)	(0.377)	(0.416)
Log Population		-1.807 (1.059)	-2.165** (0.979)	-2.551* (1.251)
Urban Population Share		-0.009 (0.065)	-0.036 (0.103)	0.087* (0.043)
year=1997		0.000 (.)		
year=1999		0.116* (0.062)		
year=2001		0.207*** (0.067)		
Constant	65.736*** (0.870)	89.769*** (18.063)	96.964*** (14.464)	95.615*** (21.483)
Observations	675	648	378	483
R^2	0.416	0.467	0.227	0.645

Standard errors in parentheses

All robustness models include year fixed effects shown explicitly,

Clustered standard errors at country level.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$